SSC , carcinoma insitu epidermal tumor

Dr eman gomaa

Consultant dermatologist

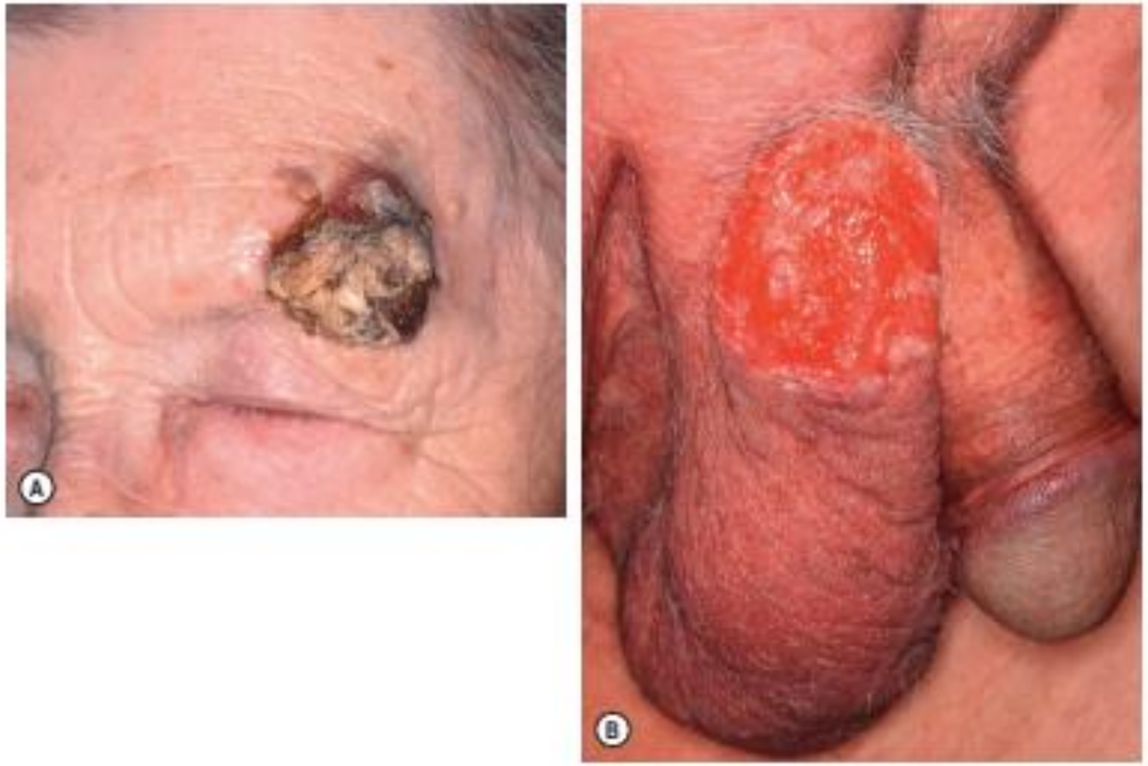
Invasive scc

- Malignant tumor of kcs and its appendegs
- 2nd most common skin cancer
- White population
- male

- Erythamatous to skin colored papulo-nodule >>>may be
- Exophytic or papillomatous plaque >>>> hperkaratotic
- Site :
  - Sun damged skin >>>dorsum of hand , forearm
  - Bald scalp , lower lip , pinna

Fig. 108.7 Clinical spectrum of cutaneous squamous cell carcinoma (SCC). **A** Eroded and keratotic nodule that developed rapidly at the site of trauma on the shin. **B** Large fungating nodule on the dorsum of the hand. **C** Multiple eroded superficial SCCs in association with actinic damage and sites of previous treatment on the cheek and neck of an elderly man. **D** Eroded, slightly vegetating, thick plaque arising within lichen sclerosis of the vulva. *A, Courtesy, Jean L Bolognia, MD.*

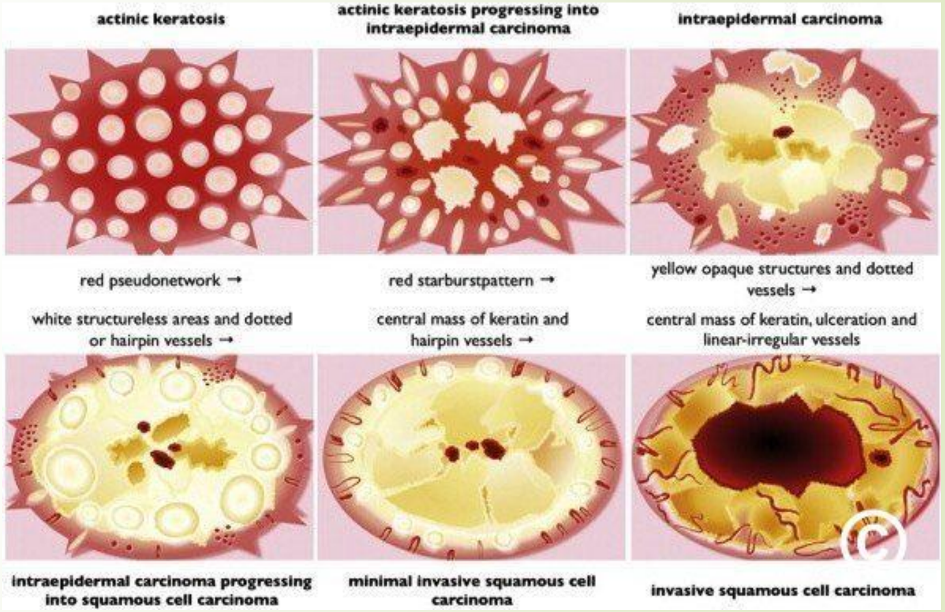
Fig. 108.8 Squamous cell carcinoma (SCC) of the lower lip. A Extensive hyperkeratosis and leukoplakia of the lower vermilion lip; histopathologically, the SCC was superficially invasive and well-differentiated. B Verrucous and eroded nodule on the lower vermilion lip in a heavy smoker.



eFig. 108.9 Clinical spectrum of cutaneous squamous cell carcinoma (SCC). A A large keratotic nodule of the supraorbital region in an elderly woman; note the coarse wrinkling and solar elastosis of the face. B Firm, eroded, red plaque on the scrotum; histopathologically, the SCC was moderately differentiated.

Dermoscopy

- Central mass of keratin , ulceration >>>surrounded by B.V
  - Atypical vascular pattern >>> linear , dotted , hairpin
  - Vessels surrounded by vascular halo

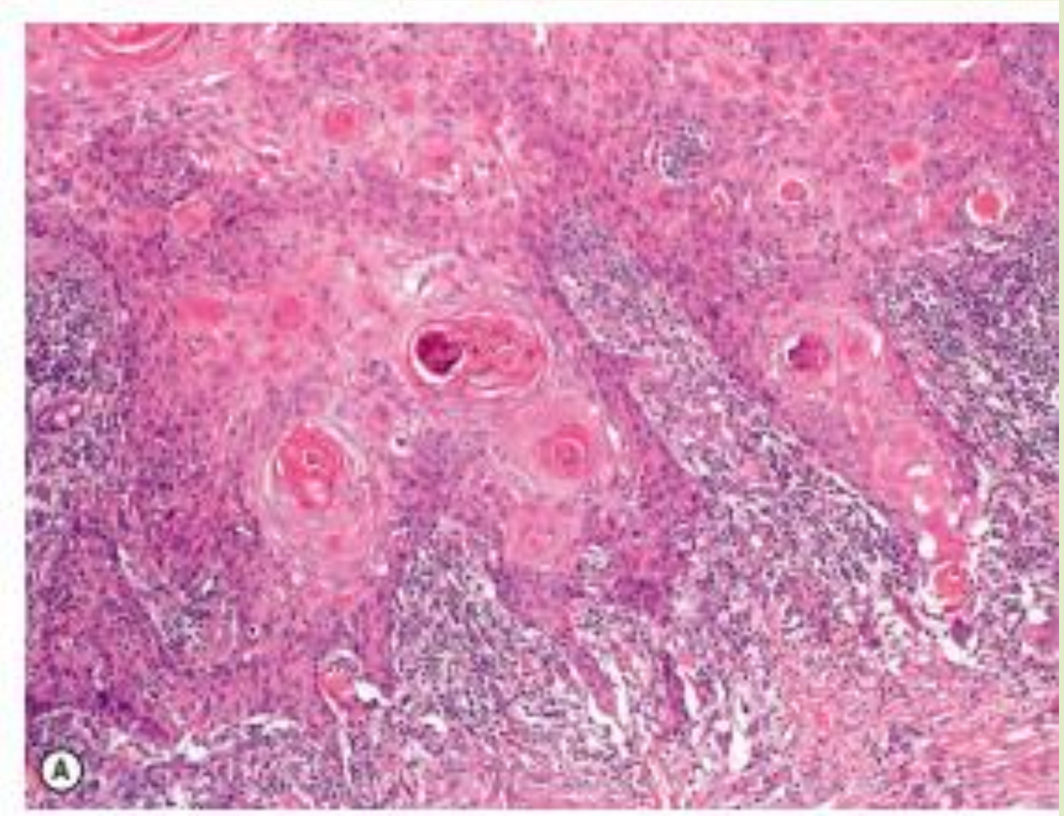


H.P

1. Well differentiated scc :

Glassy bright eosinophilic kcs

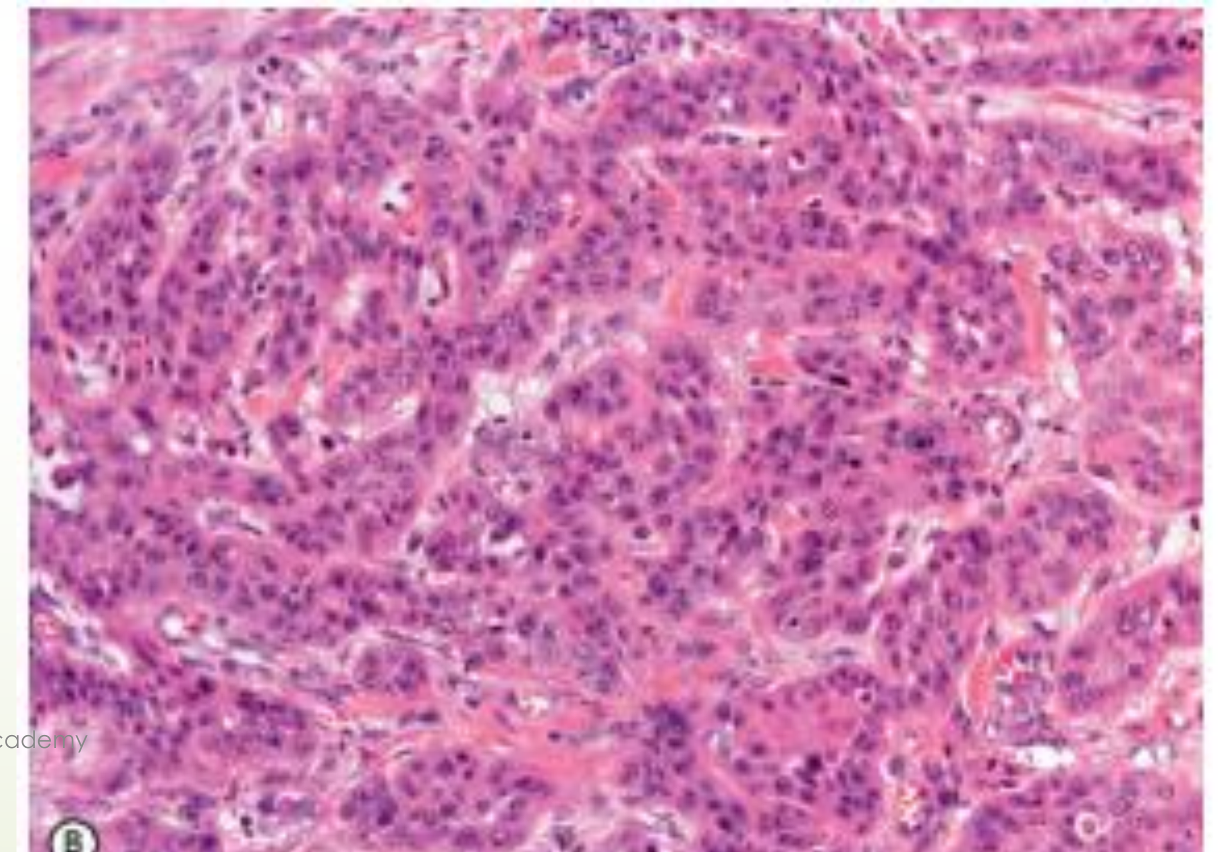
Horn (keratin pearls : concentric layer of squamous cell with gradual increase keratinization toward center



1. Poorly differentiated scc :

Strands or cords of epithelial cells with >>>> neuclear atypia

Some time spindle cell



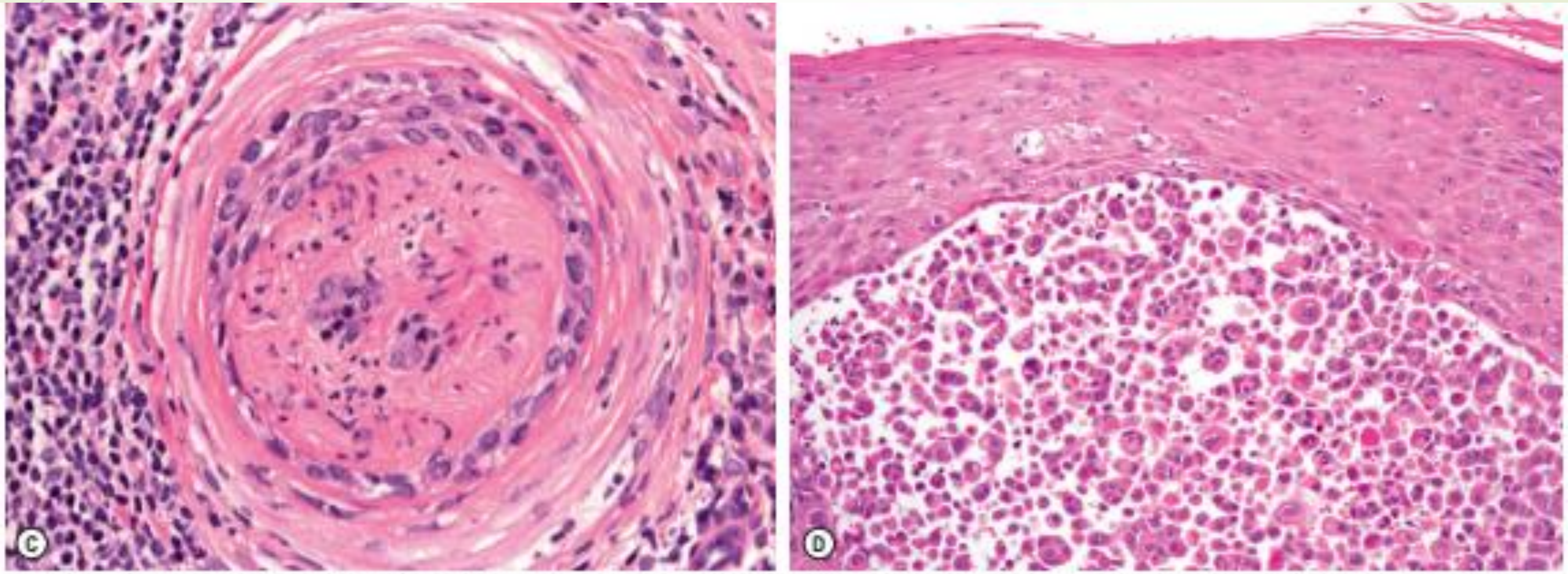


Fig. 108.13 Squamous cell carcinoma (SCC): range of histopathologic findings. A Well-differentiated SCC with numerous horn pearls (eosinophilic parakeratotic keratinization). **B** Poorly differentiated SCC with strands and cords of epithelial cells with prominent nuclear atypia and no signs of keratinization. **C** Perineural growth and neurotropism of a poorly differentiated SCC. **D** Acantholytic SCC characterized by epithelial aggregations composed of atypical keratinocytes with acantholysis. *Courtesy, Lorenzo Cerroni, MD.*

Staging

RISK FACTORS FOR METASTASIS OF INVASIVE SQUAMOUS CELL CARCINOMA
Tumor thickness: >2 mm (high risk: tumor thickness >6 mm)
Diameter: >2 cm
Location: ear, lips, mucosae including tongue, vulva, penis (perineural growth may be an additional risk factor in these locations)
Arising within a scar (e.g. burn, radiation)
Histopathologic features: poorly differentiated or undifferentiated, acantholytic*, developing within Bowen disease
Immunosuppression
*Recently questioned[20a].

STAGING OF CUTANEOUS SQUAMOUS CELL CARCINOMA (SCC) OF THE HEAD AND NECK			
T, N, M			
Primary tumor (T)			
TX	Primary tumor cannot be identified		
Tis	Carcinoma *in situ*		
T1	Tumor <2 cm in greatest dimension		
T2	Tumor 2 cm or larger, but smaller than 4 cm in greatest dimension		
T3	Tumor 4 cm or larger in maximum dimension or minor bone erosion or perineural invasion or deep invasion*		
T4	Tumor with gross cortical bone/marrow invasion (4a); skull base invasion and/or skull base foramen involvement (4b)		
Regional lymph nodes (clinical) (cN)			
NX	Regional lymph nodes cannot be assessed		
N0	No regional lymph node metastasis		
N1	Metastasis in a single ipsilateral lymph node, ≤3 cm in greatest dimension and ENE(–)		
N2	Metastasis in a single ipsilateral lymph node, >3 cm but ≤6 cm in greatest dimension and ENE(–) (2a); or in multiple ipsilateral lymph nodes, none >6 cm in greatest dimension and ENE(–) (2b); or in bilateral or contralateral lymph nodes, none >6 cm in greatest dimension and ENE(–) (2c)		
N3	Metastasis in a lymph node, >6 cm in greatest dimension and ENE(–) (3a); or metastasis in any node(s) and ENE(+) (3b)		
Distant metastasis (M)			
M0	No distant metastasis		
M1	Distant metastasis		
Stage			
	T	N	M
0	Tis	N0	M0
I	T1	N0	M0
II	T2	N0	M0
III	T3	N0	M0
	T1	N1	M0
	T2	N1	M0
	T3	N1	M0
IV	T1	N2	M0
	T2	N2	M0
	T3	N2	M0
	T Any	N3	M0
	T4	N Any	M0
	T Any	N Any	M1

*[illegible]

- Treatment of SCC
  - surgical
    - Excesion
    - destruction
  - Non surgical
    - Photodynamic therapy
    - Radiotherapy

12

Verrucous scc

- Clinicopathological variant of scc
- Well differentiated low grad indlent malignancy
- Middle aged
- HPV
- 3tpes :
  - Oral florid papillomatosis
  - Buschke lowenstien (gaint codyloma accumenata)
  - Epithelima cuniculatum

Fig. 108.10 Verrucous carcinoma. A Longstanding large nodule on the plantar surface with a rabbit burrow-like appearance; such a tumor is also referred to as an epithelioma cuniculatum. B Keratotic and ulcerated plaque on the ventromedial aspect of the great toe. C A classic location in an amputation stump. In general, these well-differentiated SCCs enlarge slowly.


Logical Images, Inc.

- Larg exophytic with paplomatus or verrucous surfase
- H.P :
  - Same scc
  - Pushing border not infiltirating
  - Minimal atypia
- Prognosis
  - Gradual penetration , destruction
  - No metastasis

Keratoacanthoma

- Rapidly growing ,self healing ,
- benign keratinizing Squamus cell carcinoma
- (Well differentiated scc)
- Self limited within 6 months

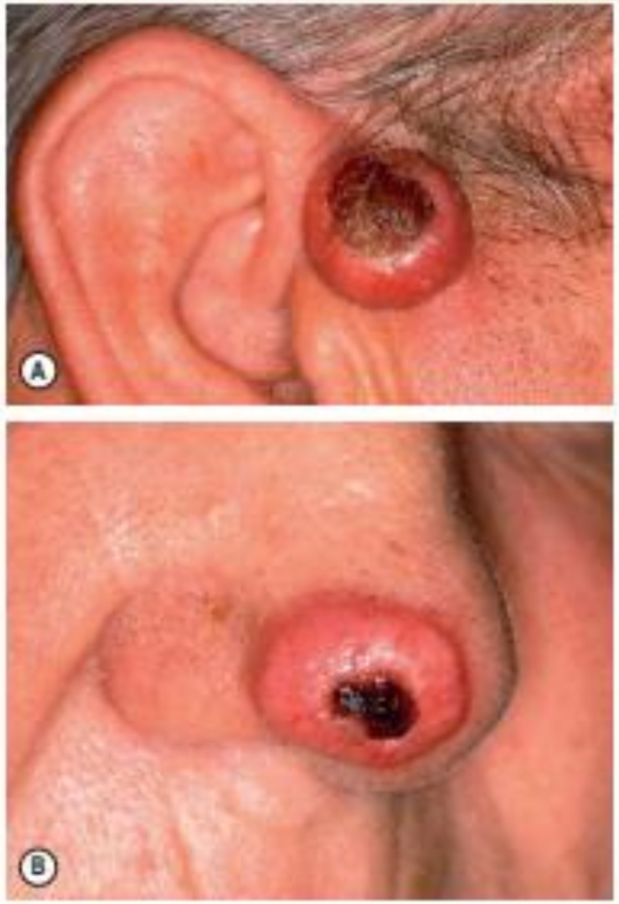


Fig. 108 spectru keratoa A,B Rap erythem craterifo a rolled central Progres expansi involuti atrophy keratoa centrifu margina keratoa yellow– history

Types of KA

Solitary KA

Common , middle life , male , white population
Head , neck
Solitary , rapidly enlarging
>>papule>> nodule >>
keratin filled crater

TREATMENT :
No tt or excisional biopsy
Superficial destructive method

Varients : KA centifugum , marginatum , grouped Ka , subungual KA, gaint KA

Multiple KA

Chemical , immunosuppression, BRAF inh. , HPV

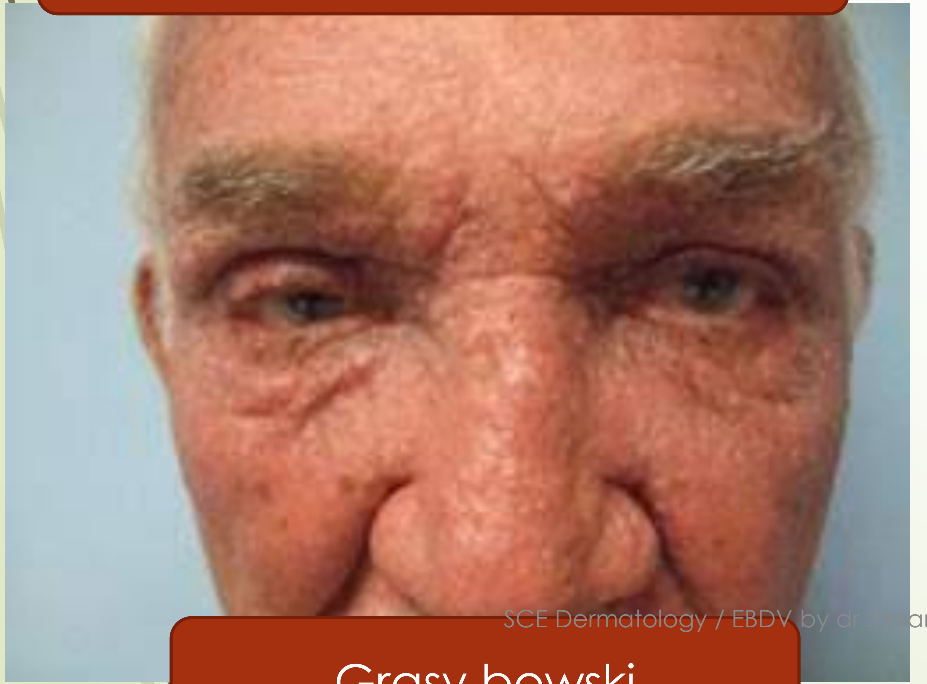
Ferguson smith

Familial multiple keratoacathoma
AD –TGFBR1, TGFB
Sun exposed , 3rd decade
Spontaneusly regression
No tt
Oral retenoid

Grasy bowski

Generalized KA
Unkown
Thousand like miliaria
Scaring , ectropian , mask like face
Slowly resolve
Tt:
Acitretine 10 -25mg

Freguson smith



Grasy bowski

H.P OF KA

- Lesion totally excised at lower magnification
- Volcano like architecher >>>>
- with keratin filled invagination of epidermis
- Well differentialted kcs , keratin pearls
- Minimal atypia

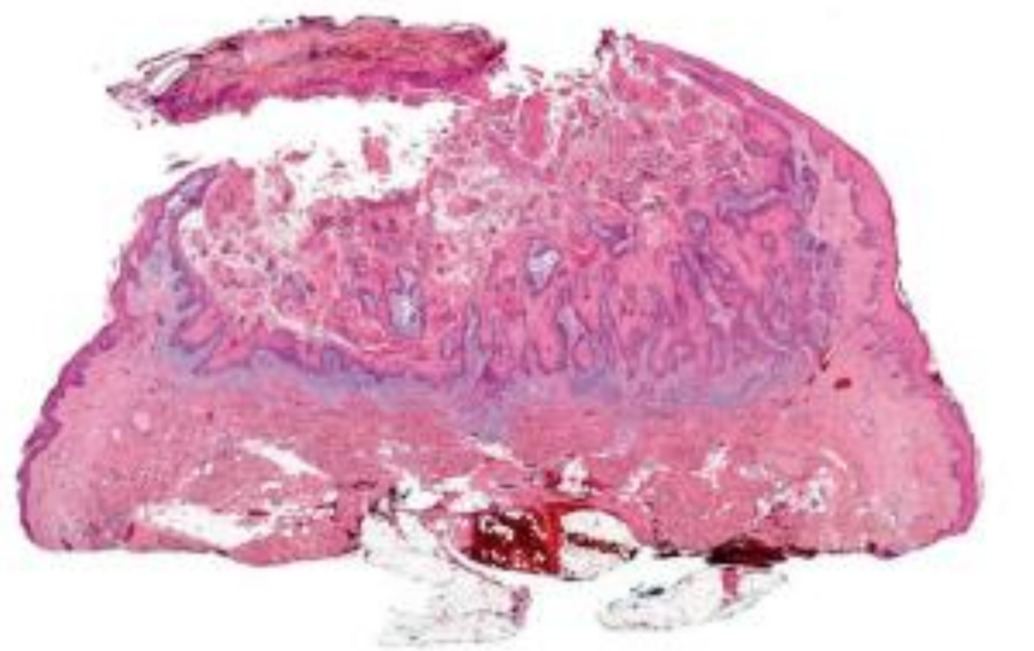
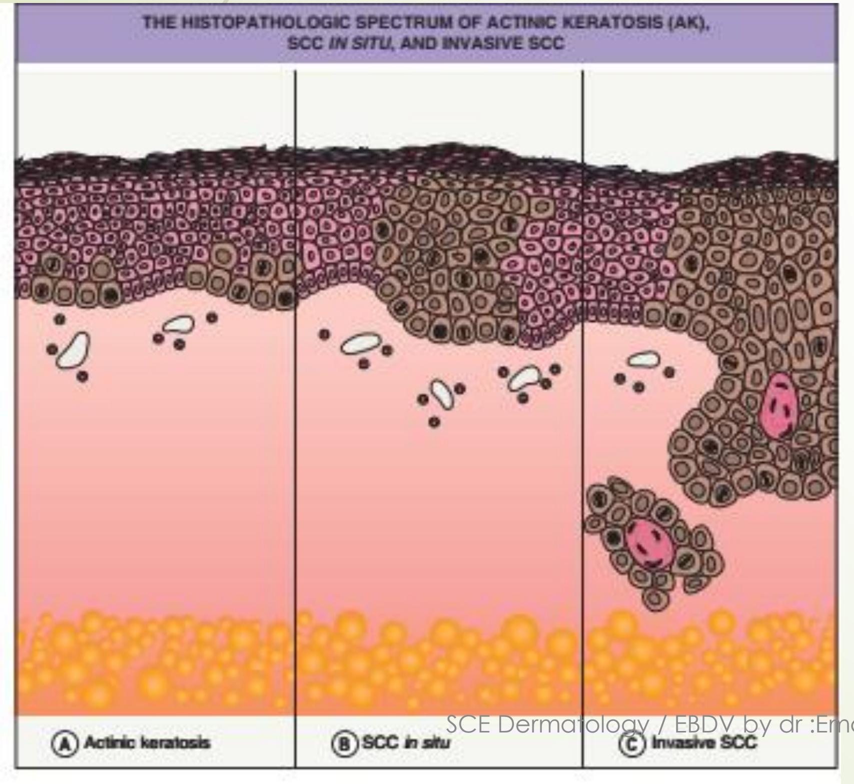


Fig. 108.14 Histopathology of keratoacanthoma. Scanning magnification of a stereotypical keratoacanthoma with a keratin-filled crater encompassed by epithelial lips. Courtesy Lorenzo Cerroni, MD.

Carcinoma insitu



- Actinic keratosis
- Bown disease
- Pagets disease

Psoraisiform or eczematous patch

Erythamatous patch covered with scales

Actinic keratosis

- Very common
- Old wight male ,sun exposur , immunosuppression
- Rough erythamatuos papules >>>with white yellow scal
- Sun damged skin >>> head , neck

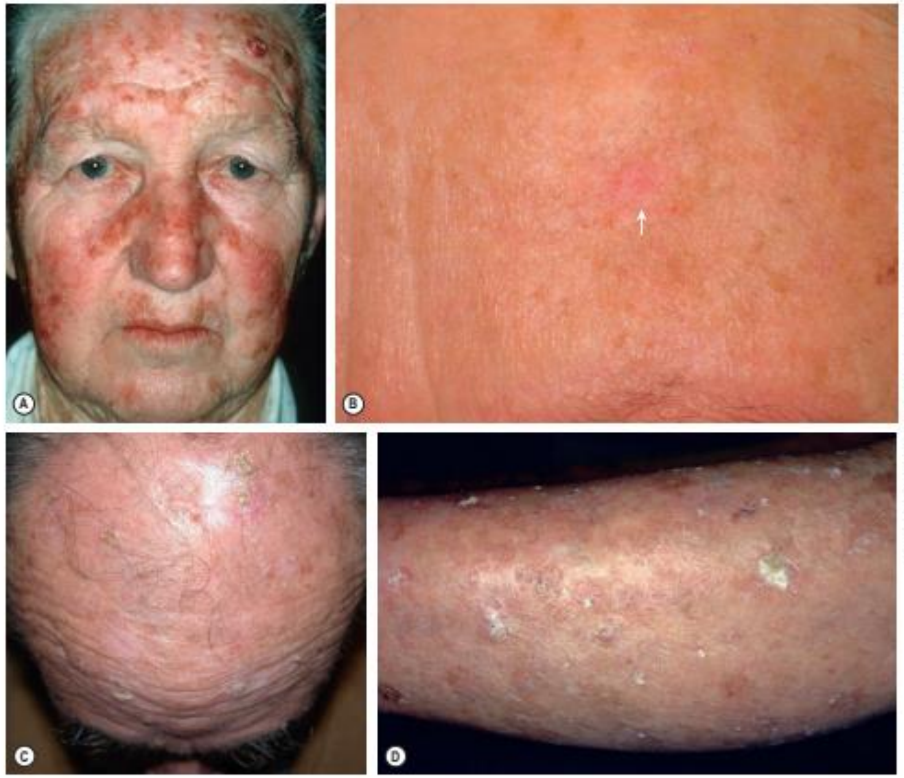
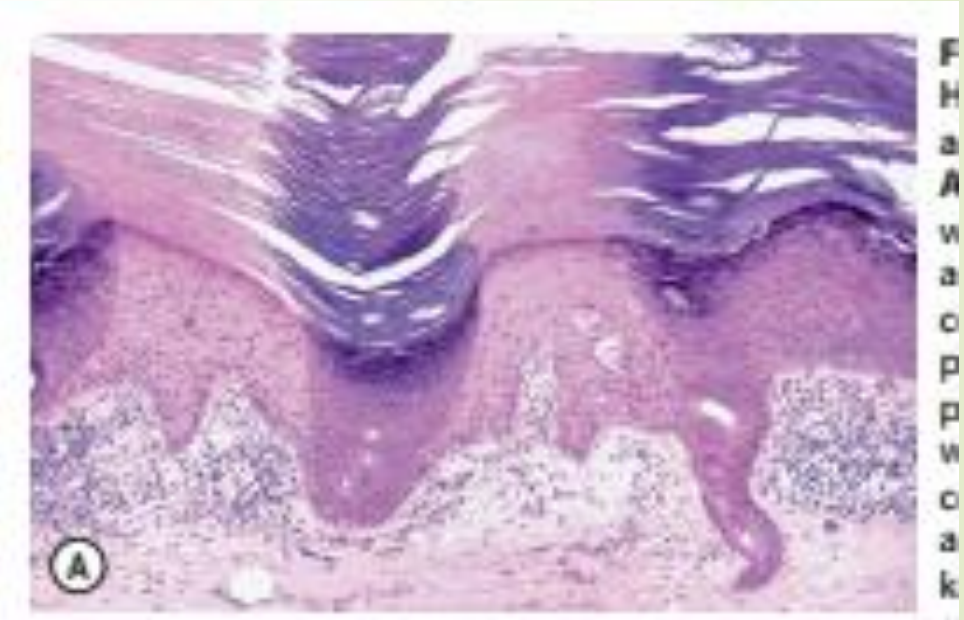
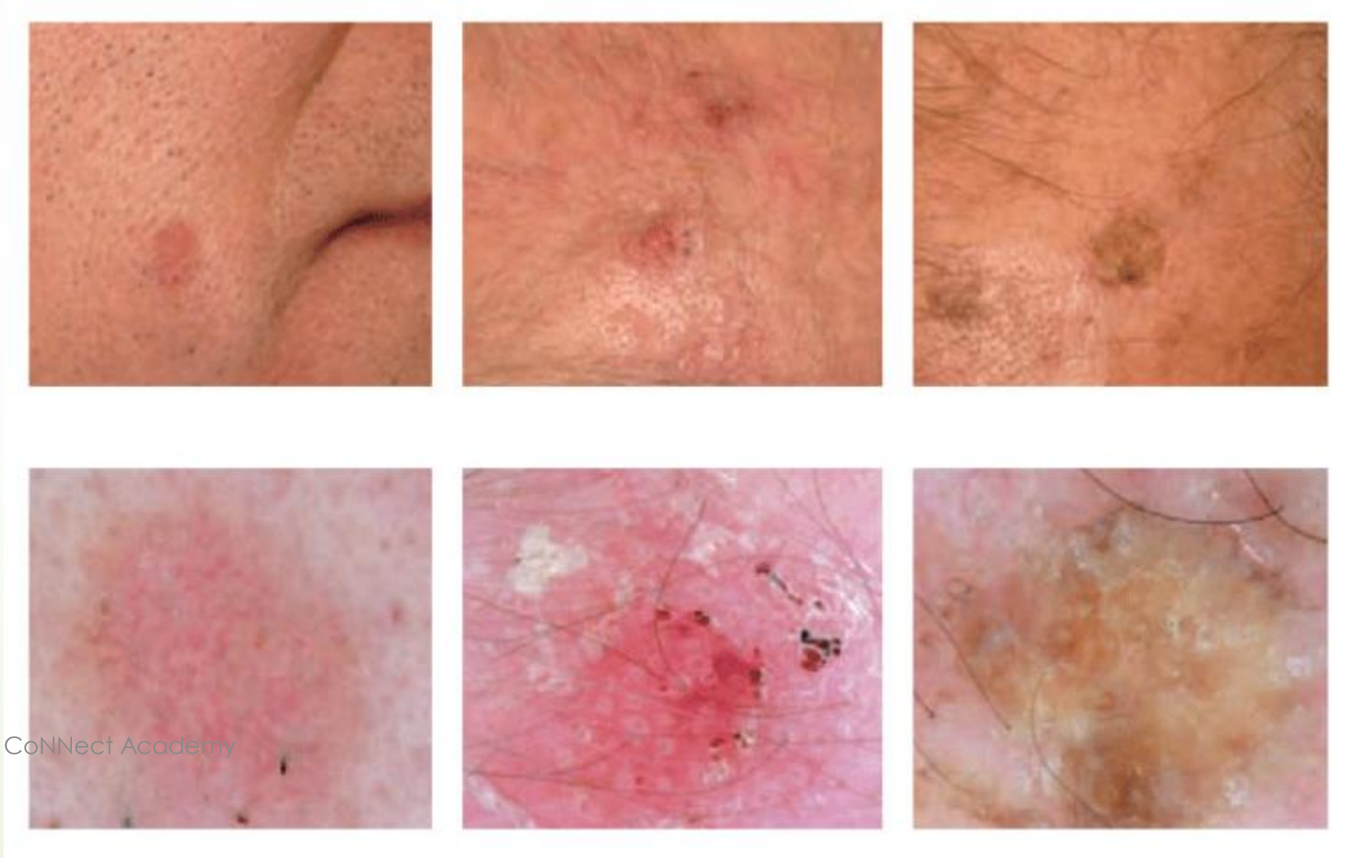


Fig. 108.2 Actinic keratoses (AKs). **A** Multiple AKs on the face of an elderly woman with fair complexion, blue eyes, and moderate to severe photodamage; the AKs vary in size from a few millimeters to over a centimeter. On the left forehead, the red nodule with slight scale-crust represents a well-differentiated SCC. **B** Pink-colored atrophic AK with minimal scale on the forehead. **C** Multiple AKs on the bald scalp, some of which are hyperkeratotic; the area of hypopigmentation represents a site of previous treatment and due to the field defect, recurrence at the edge is commonly observed. **D** Multiple, large hypertrophic AKs on the shin of an elderly woman; note the thick scale. *B, Courtesy, Iris Zalaudek, MD; D, Courtesy, Jean L Bolognia, MD.*

- H.P :
  - Altarnating ortho, parakeratosis
    - Pink , blue sign (flag sign)
  - Atypical kcs >>>partial thickness >>>
  - Limited to lower epidermis
  - Spared adenexal epithelium
  - Dermis :inflame- cells , solar elastosis



- Dermoscopy :
- Erythema , plugs , scale
- (seeds of strawberry



- Clinical varients :
  - Hypertrophic
  - Atrophic
  - Pigmented
  - Lichenoid
  - Actenic chelitis
- Course :spontaneous regress or evolve to scc
-

Fig. 108.4 Actinic cheilitis. Erythema and scale of the entire lower vermilion lip

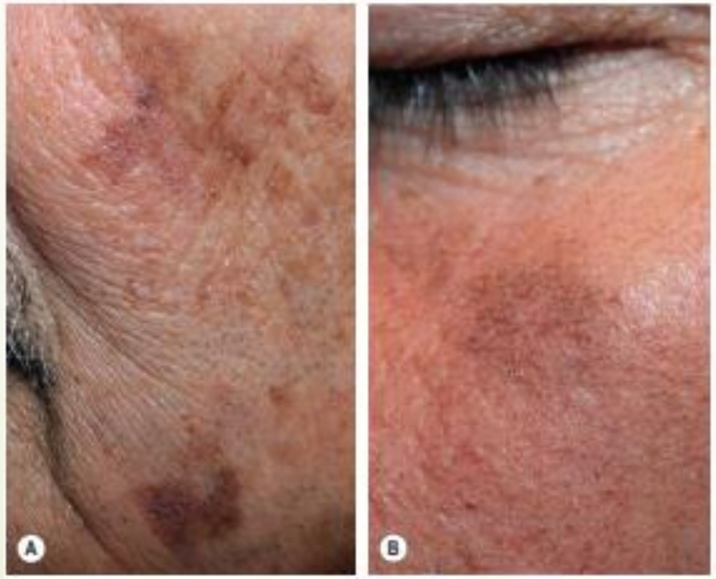
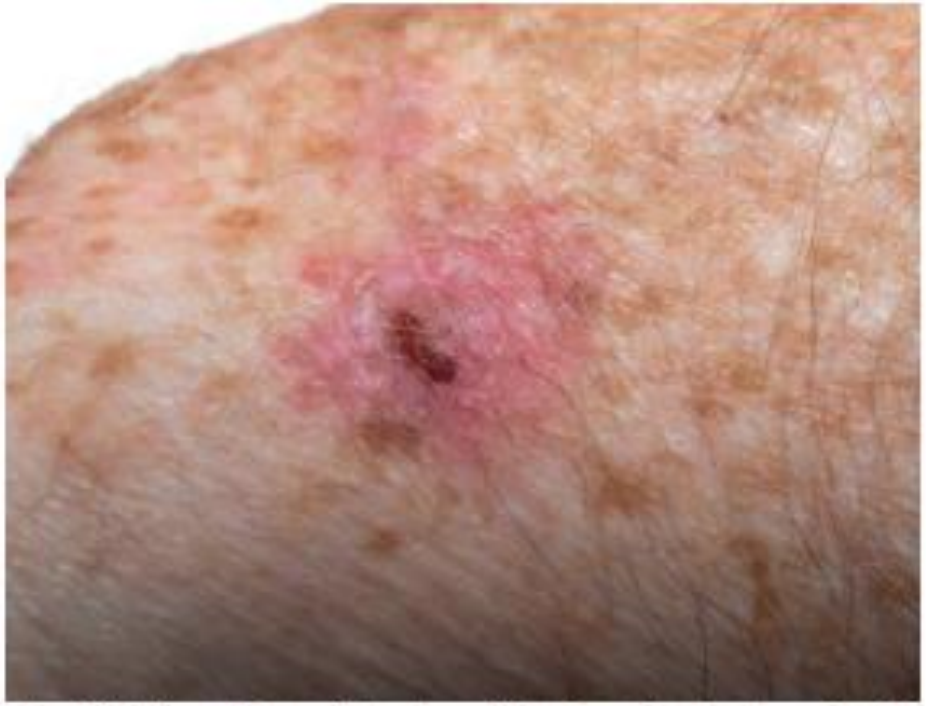
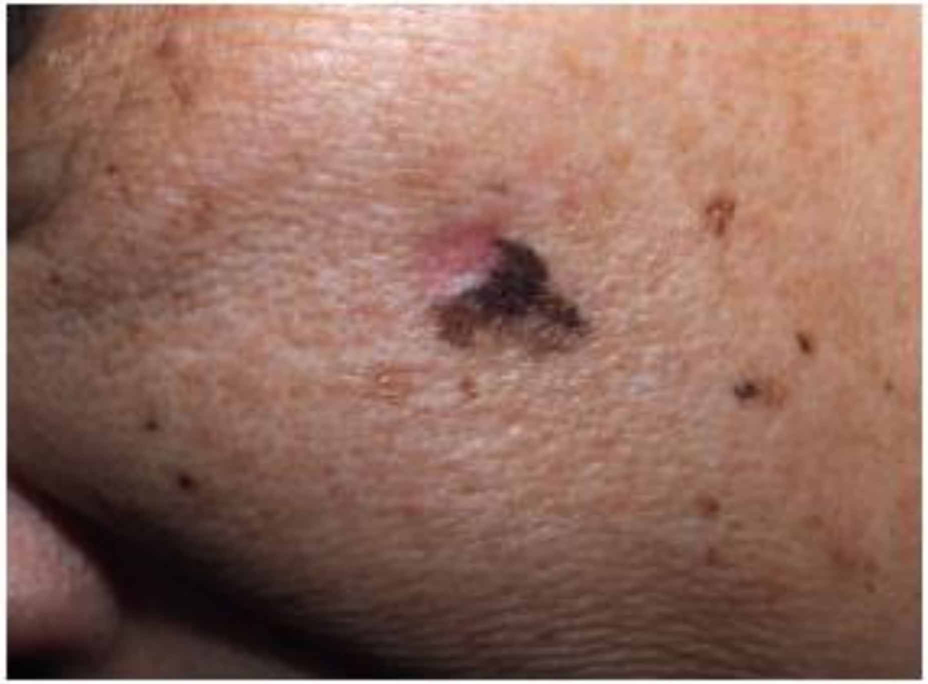


Fig. 108.3 Pigmented actinic keratoses. A,B The hyperpigmentation can have a reticulated appearance and there may be associated scale but an absence of



eFig. 108.7 **Bowen disease.** Pink plaque with white scale and a hemorrhagic crust arising within photodamaged skin. *Courtesy, Kalman Watsky, MD.*



eFig. 108.8 **Pigmented Bowen disease.** This lesion may be diagnosed as a cutaneous melanoma clinically. The pink depressed area is a biopsy site. *Courtesy, Kalman Watsky, MD.*

- Treatment :
  - Superficial destructive method
  - Topical chemotherapy :
    - 5fu , imiquimod ,
  - Contraindicated NB , puva

- DD :erythamateus scaly patch

BOWENS DISEASE

- Arised denovo or from prexisting AK
- Old white women
- Sun exposed , sun protected
- Sharp demarcated but irregular erythematous scaly solitary patch
- Mucosal , genital skin >>>high invasion , recurrence
- Nail (subangual , periangual >>>hpv16

Fig. 108.5 Squamous cell carcinomas *in situ*, Bowen disease type. A Scaly red plaque on the chest with skip areas and background photodamage. **B** Larger broken-up pink plaque with scale-crust in the pubic region, a sun-protected site. This type of lesion is often misdiagnosed as dermatitis or psoriasis and treated with topical corticosteroids. **C** Bright red, well-demarcated plaque on the proximal nail fold with associated horizontal nail ridging; the possibility of HPV infection needs to be considered. **D** Dermoscopic findings of tiny dotted vessels in the upper half of the lesion combined with superficial scales. **E** Extensive involvement of the finger which was misdiagnosed clinically as an inflammatory dermatosis and treated for years with corticosteroid creams. *B, Courtesy Kalman Watsky, MD; D, Courtesy, Iris Zalaudek, MD.*

- H.P
  - Diffuse parakeratosis
  - Full thickness atypia >>(**wind blown**)
  - Affect adenex

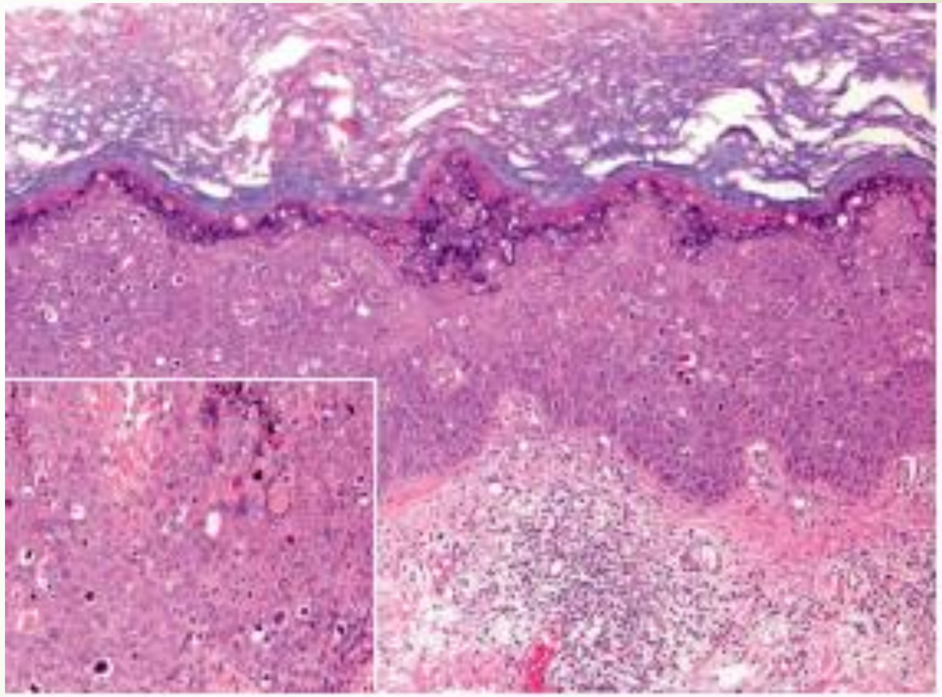
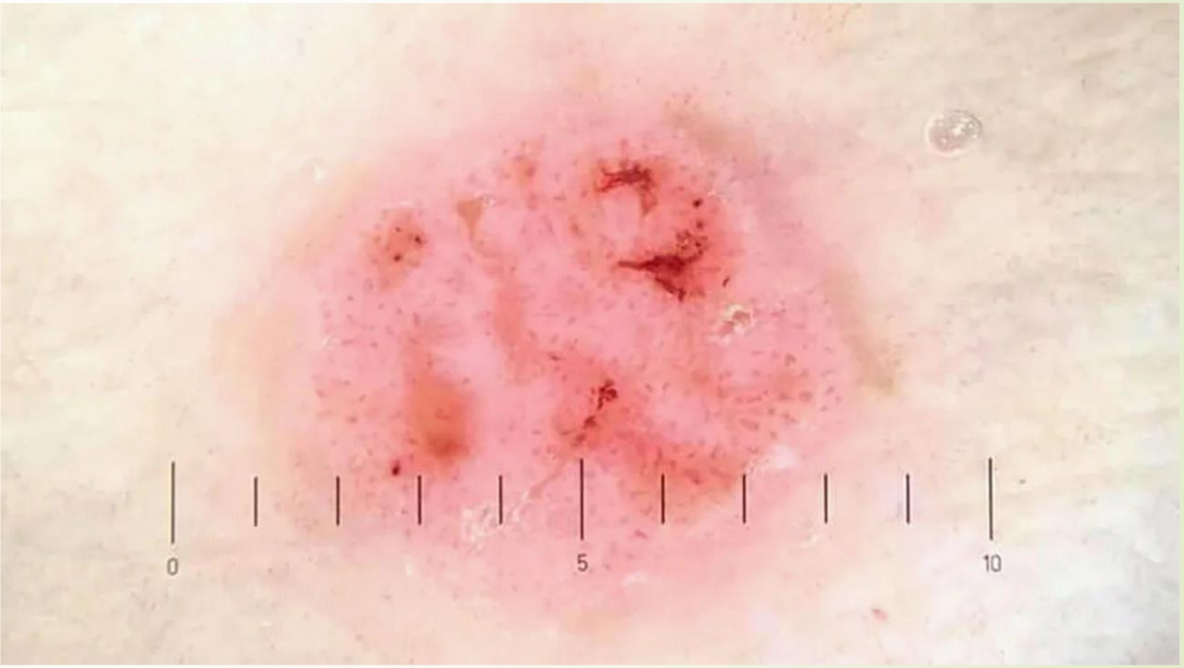


Fig. 108.12 Histopathology of Bowen disease (squamous cell carcinoma *in situ*). Irregular epidermal hyperplasia with atypical pleomorphic keratinocytes throughout the entire epidermal thickness. Note several necrotic keratinocytes (inset). *Courtesy Lorenzo Cerroni, MD.*

- Clinical variants :
  - **Erythroplasia of queyrate**
  - **Bowenoid papulosis**
  - Arsinic
  - Pigmented
  - Verrucous

Fig. 108.6 Squamous cell carcinoma *in situ*, erythroplasia of Queyrat type. Large, eroded erythematous plaque with well-demarcated borders. The lesion began on the shaft of the penis.

- Dermoscopy :
  - Glomerular vessels
  - Erythema , keratosis

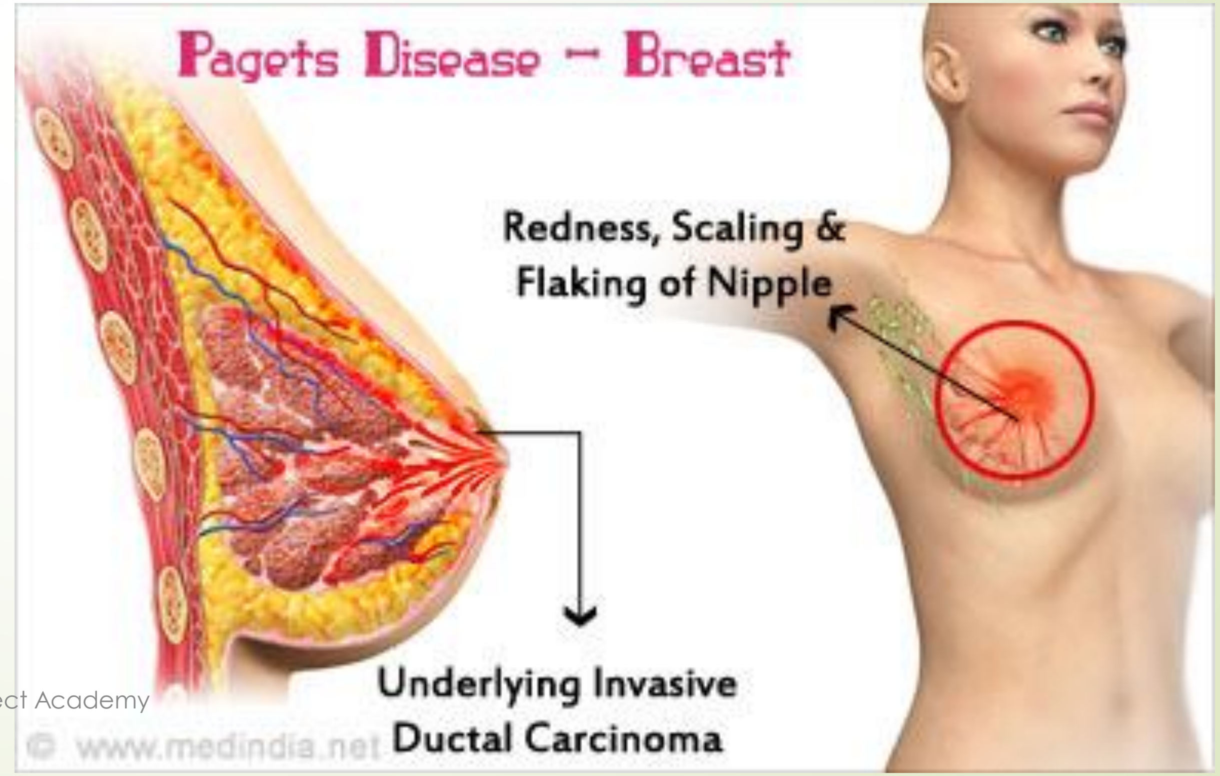


Pagets disease

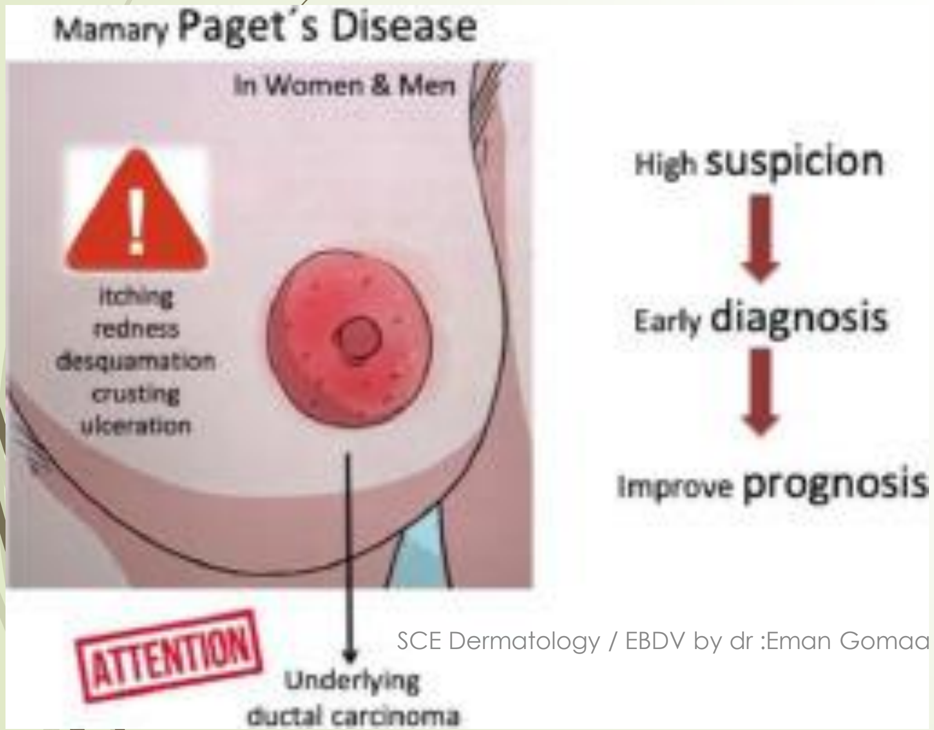
- Mammary pagets
- Extramammary pagets

Mammary pagets

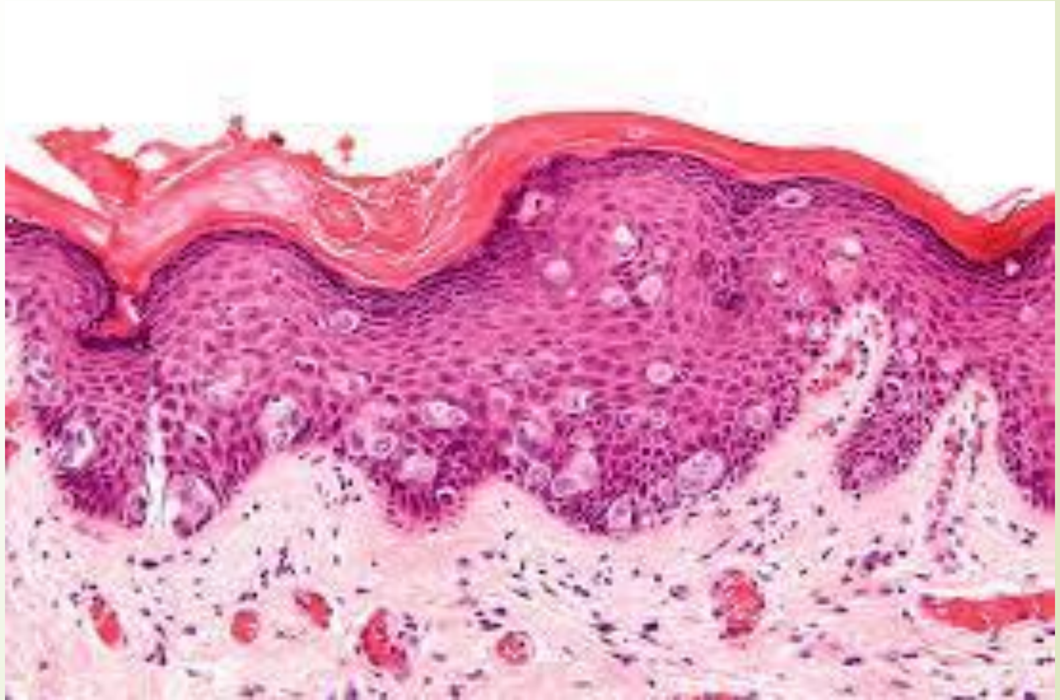
- Def :invasion of epidermis by malignant cell
- from intraductal carcinoma
- Uncommon 5th-6th decade



- c/p:
  - Unilateral scaly erythematous patch or plaque >>
  - Early >>eczematous >>late >>sharp demarcated slightly raised edge
  - Itching, pricking burning at nipple , areola



- H.P:
  - Epidermis :hyperkeratosis , parakeratosis
  - Paget cells between spinous cell l
  - (large rounded ,clear cell surrounded by halo
  - PAS+ve , diastase resistance



- Immunohisteochemestry :
  - CEA, EMA

- TREATMENT :
  - Mammogram , ultrasound
  - Surgry , radiotherapy
  - Sentinel L.N biopsy

Extra mammary paget

- Elderly , Caucasian women
- Aetiopathogenesis:
  1. 1ry :
     - Intaepithelial adenocarcinoma (apocrine origen)
       1. Vulva >>women
       2. Perianal >>men
  2. 2ry :
     - Pagetoid spread from
       1. adenexal adenocarcinom
       2. Viseral malignancy

- Sharply demarcated erytham-scaly plaque
- **(strawberries in cream)**

e 1: Clinical presentation of the well-demarked

porokeratosis

- Clonal expansion of kcs >>>>
- differentiat abnormaly not truly neoplasm
- Benign , localized , superficial
- **aetiopathogenesis:**
  - Genatic
  - Triggering factor
    - Sun exposur
    - HPV
    - immunological

Types of porokeratosis

porokeratosis	
Classic of mibilli	DSAP

LINEAR	Punctate

GAINT	
PPPD	Porokeratosis ptychtropica

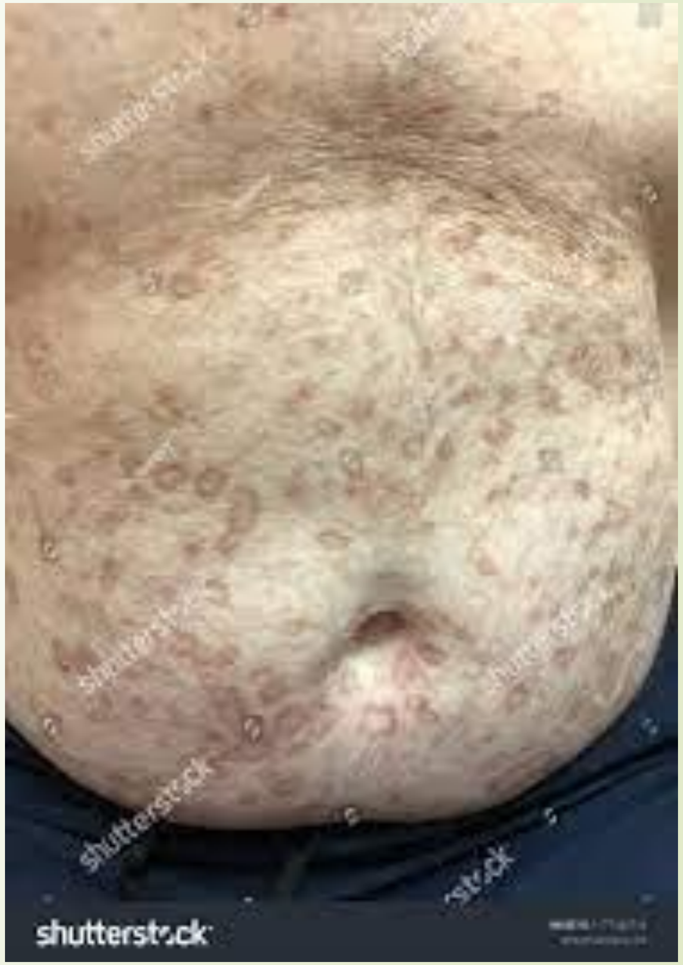
Fig. 109.8 Porokeratosis – clinical variants. A Porokeratosis of Mibelli on the hand of a child. **B** Actinic porokeratosis in a renal transplant patient with significant solar damage. Note the narrow, elevated rim. **C, D** Multiple lesions of disseminated superficial actinic porokeratosis (DSAP) with obvious peripheral keratotic rims; lesions can range in color from light brown to pink. **E** Several streaks of linear porokeratosis on the lower extremity. **F** Multiple keratotic papules of the plantar surface due to punctate porokeratosis. **G** Porokeratosis ptychotropica with pink to brown papules localized to the intergluteal cleft and buttocks. *B, C, D, Courtesy, Jean L Bolognia, MD. F, Courtesy, Kalman Watsky, MD; G, Courtesy, César Cosme Álvarez Cuesta, MD.*

CLINICAL VARIANTS OF POROKERATOSIS	
Variant	Clinical features
Porokeratosis of Mibelli	Plaque that arises during infancy or childhood, usually on a distal extremity; often several cm in diameter
Disseminated superficial actinic porokeratosis (DSAP)	Pink to brown papules and plaques with peripheral scale; arise in sun-exposed sites, especially the extensor forearms and shins; usually measure from a few mm to 1 cm in diameter; in some patients, autosomal dominant inheritance
Linear porokeratosis	Streaks along the lines of Blaschko; arise during infancy or childhood; risk of development of squamous cell carcinoma
Punctate porokeratosis	Palmoplantar papules that measure 1–2 mm in diameter; arise during adolescence or adulthood
Porokeratosis palmaris et plantaris et disseminata (PPPD)	Palmoplantar papules in addition to involvement of the trunk, extremities, and even mucous membranes; onset during childhood or adolescence
Porokeratosis ptychotropica	Red to brown papules or plaques in the intergluteal cleft and on the buttocks; there may be coalescence of lesions centrally with scattered papules peripherally
Eruptive disseminated porokeratosis	Abrupt onset of numerous, widespread, inflamed keratoses; may be a sign of an associated malignancy in up to ~30% of patients

- **Classic porokeratosis of mibelli**
  - Rare
  - Boy , infancy , childhood
  - Keratotic papule >>>plaque >>>
  - Raised , well demarcated , keratotic thready border >>
  - With longitudinal furrow
  - Extremities

Dissemenated superficial actinic porokeratosis

- (DSAP)
- HCV , SLE
  - The most common
  - Women 3rd – 4th decade
  - Caucasian sun exposur
  - Thin keratotic papules >>thin elevated furrowed keratotic rim
  - Bilateral symmetrical
  - Sun exposed , extremities
  - Spare face



LINEAR porokeratosis

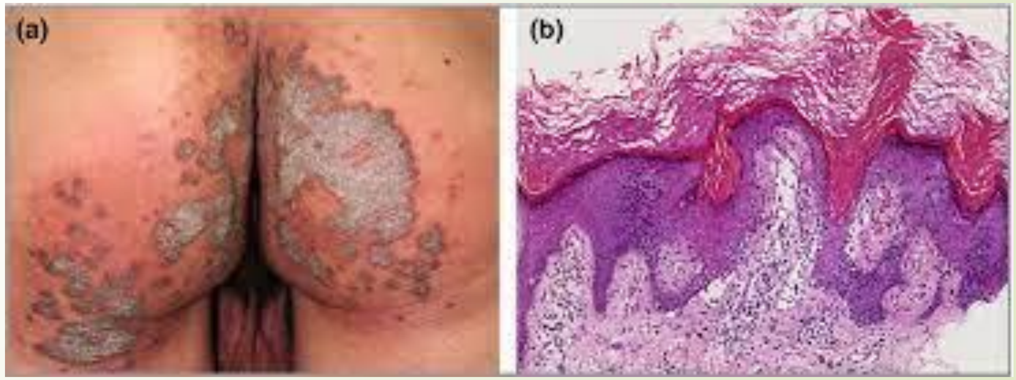
Infancy , childhood
Blashko lines
Extremeties
scc

Punctate

Difficult to diagnosis
Adolscence ,
adulthood
Seed like
Palm , sole

PPPD
Uncommon
Child , adolescence
Male
AD

Porokeratosis
ptychotropica
Verrucous
Intergluteal cleft
Buttocks
Multiple
Cornoid lamella



Histopathology
Coronoid lamella

1. Column of parakeratosis.
2. Hypogranulosis.
3. Dyskeratosis.
4. Vacuolar degeneration.
5. Lymphocytic infiltration.

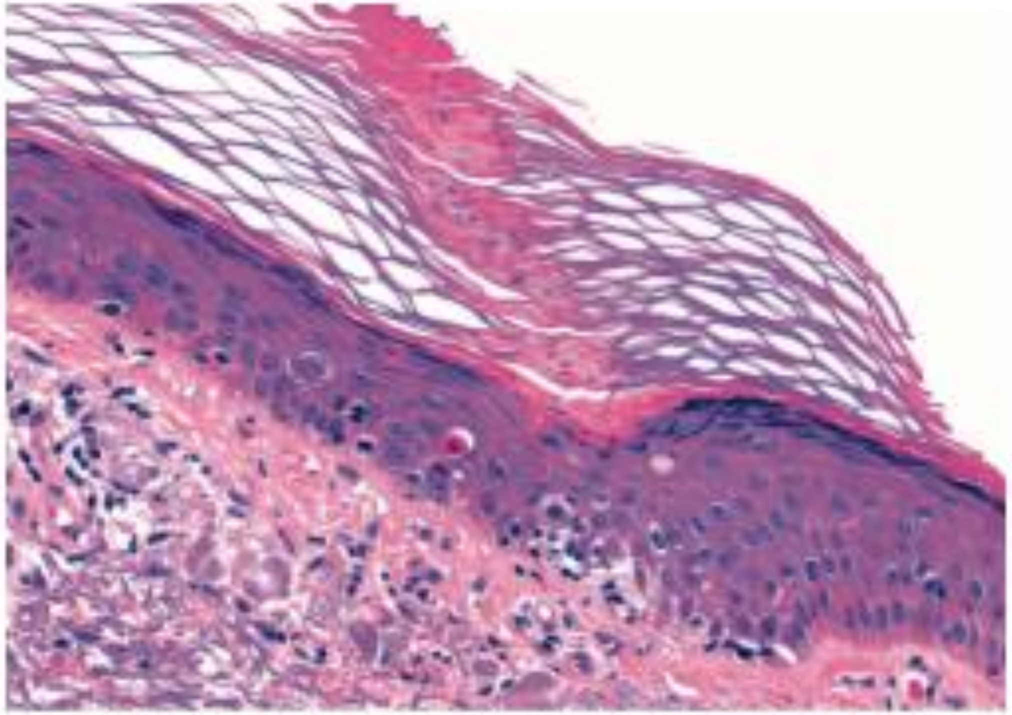


Fig. 109.9 Porokeratosis – histologic features. The cornoid lamella, a narrow column of parakeratotic cells, is the typical histologic feature of porokeratosis. Note both the absence of the granular layer and the vacuolar change at the dermal–epidermal junction beneath the cornoid lamella.

Treatment

- Superficial destructive method
- Cryo, F5U & Retinoid.
- Oral acitretine.

- Epidermal Navus
  - Epidermal Verrucous Nevus
  - Ilven
  - Nevus Comedonicus

Epidermal (Verrucous) Nevus

- Epidemiology
- 1:1000
- 1^{st} year
- Equal sex
- At (shortly after) birth.

Etiology

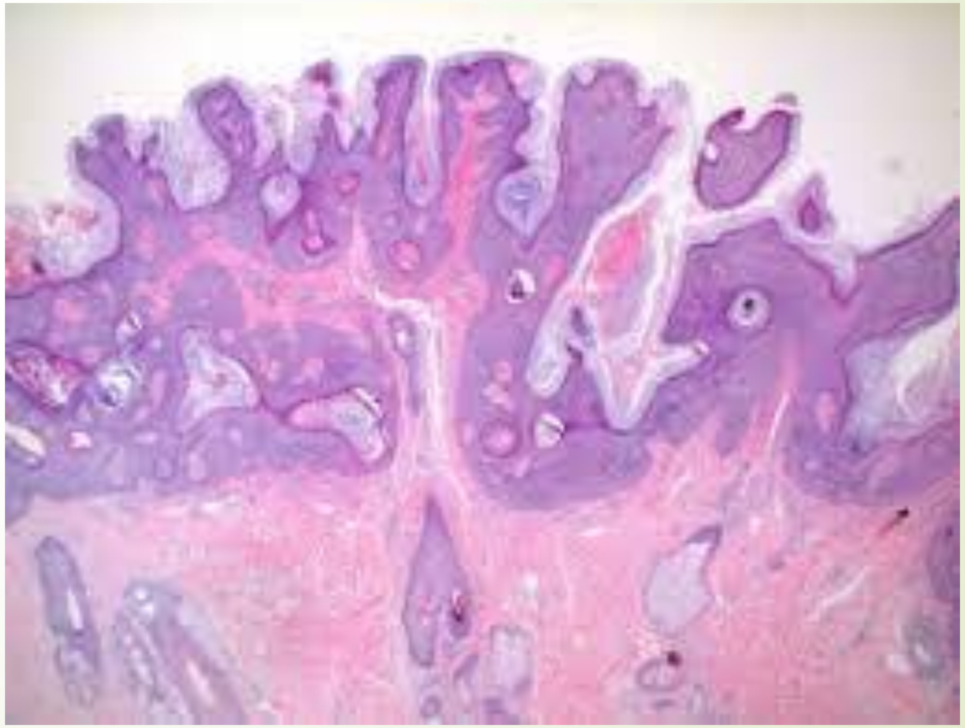
- Sporadicly
- Originate from pluripotent cells of
- basal layer of embryonic epidermis.
- N.B.:- Hamartoma → Papillary dermis

C. pic

- Single, unilateral, linear & follow blashko line.
- **Hyperpigmented – Papillomatous papule,**
- **plaque involoving trunk and neck.**
- **Variants**
- **Localized:-** Nevus unius lateralis.
- **Generalized:-** Ichthyosis hystix.
- **Epidermal nevus syndrome.**

HP

- Hyperkeratosis, focal parakeratosis, acanthosis & papillomatosis.
- Folds → acantholytic dyskeratosis.
- DD
  1. Nevus sebaceus.
  2. Lichen sreatus
  3. Overgrowth syndrome.
     1. Proteus.
     2. Cloves
     3. PTEN – hamartoma tumor syndrome.



TTT

- Full thickness surgical excision

Ilven
Dermatitic epidermal nevus

- Epidemiology:- childhood 75% < 5h.
- 4 times in girl.

❑ Etiology:-

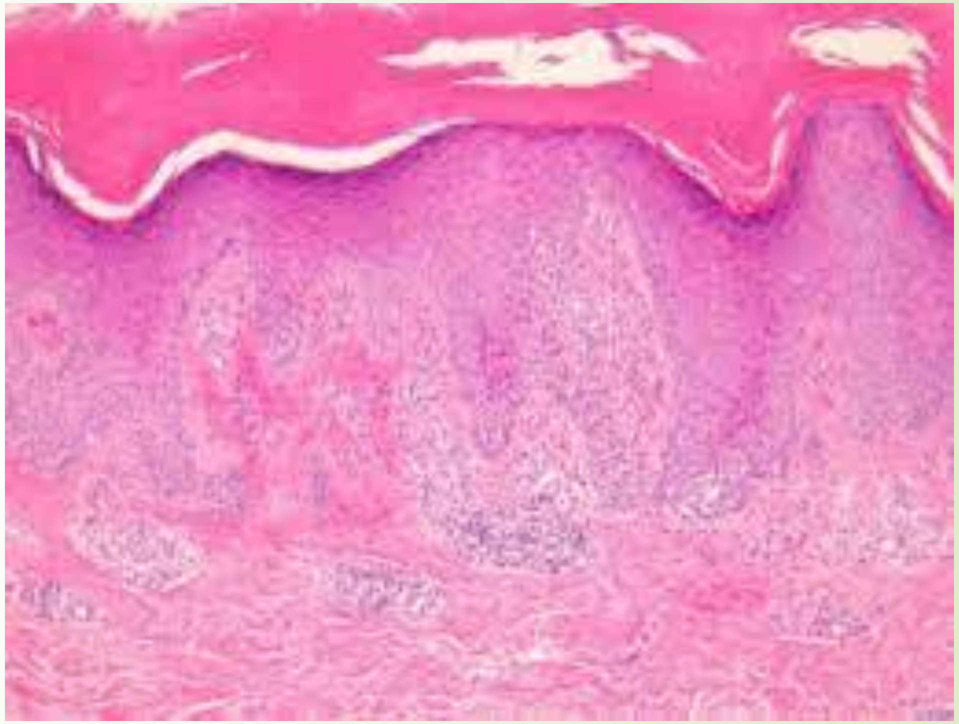
- unknown with familial predisposition.
- Relation to patients.
- KCs grow abnormality.
- Mosiac erythrokeratoderma variabilis.

C.Pic.

- Highly pruritic Psoraisiform appearance
- Scaly erythematous papules → Linear plaque
- Unilateral LL left leg
- **Association**:- Arthritis.

HP

- Psoraisiform hyperplasia.
- Ortho - parakeratosis
- Exocytosis, Spogiosis → Eczema.



- DD

- Child syndrome.
- Psoriasis with epidermal nevus.
- Lichenoid epidermal navus.

TTT

- Difficult to treat
- Surgical ttt with residual scar.
- PDL.

Nevus comedonicus

- Benign hamartoma of folliculosebaceus gland
- Dilated keratin filled commedon
- FGFR2
- Single well definedlinear streak of cluster of comedon

A 78-year-old man attended clinic for biopsy results of a 15mm lesion on his upper back. The histology report showed a moderately differentiated squamous cell cancer (depth 3 mm) arising within Bowen's disease.

What feature of the histology report is regarded as high risk for recurrence?

- Arising from Bowen's disease
- Differentiation (moderate)
- Depth (3 mm)
- Location (back)
- Size of lesion (15mm)

Arising from Bowen's disease ← Correct answer	38%
Differentiation (moderate)	19%
Depth (3 mm)	27%
Location (back)	5%
Size of lesion (15mm)	11%

Explanation:

Squamous cell cancers arising from Bowen's disease are considered high-risk.

Other high risk features include site (lip, ear, non-sun exposed sites, areas of chronic inflammation/ injury), greater than 2 cm diameter, greater than 4mm depth/ invading subcutaneous tissues, poorly differentiated/ desmoplastic/ acantholytic and spindle subtypes, perineural/ lymphatic/ vascular invasion, patients on immunosuppression, recurrent/ incompletely excised.

Reference:

- R J Motley, P W Preston, C M Lawrence. Multi-professional Guidelines for the Management of the Patient with Primary Cutaneous Squamous Cell Carcinoma. BAD 2009

A 70-year-old Caucasian man presented with a lesion on his forehead. The lesion started as a pimple a few months earlier but had been growing over the last two weeks. On cutaneous examination, a non-tender dome-shaped nodule, which is firm and bluish-red, is present over his forehead. Excisional biopsy histologically showed nests of uniform, round, blue cells, containing large basophilic nuclei with delicate granular dispersed chromatin and inconspicuous nucleoli, and minimal cytoplasm, which was positive for CK20 on immunohistochemistry.

What is the most likely diagnosis?

- Adnexal tumour
- Amelanotic melanoma
- Basal cell carcinoma
- Merkel cell carcinoma
- Small cell melanoma

Adnexal tumour		4%
Amelanotic melanoma		2%
Basal cell carcinoma		6%
Merkel cell carcinoma	← Correct answer	87%
Small cell melanoma		0%

Explanation:

Merkel cell carcinoma (MCC) of the skin is a rare, potentially aggressive, cutaneous malignancy that predominantly affects Caucasians with fair skin types and has a high tendency for recurrence and metastases.

It typically presents with a rapidly growing, painless, firm, non-tender, shiny, flesh-coloured or bluish-red, intracutaneous nodule commonly located in the head and neck region. The diagnosis of MCC requires a high index of suspicion, as it is often clinically misdiagnosed. Clinical warning signs that should prompt a biopsy are summarised in the acronym AEIOU (asymptomatic lesion, expanding rapidly, immunosuppression, age older than 50 years, lesion on UV-exposed skin). Clinically, MCC may closely mimic many benign and malignant lesions occurring on sun-exposed skin, such as basal cell carcinoma, squamous cell carcinoma, small cell melanoma, keratoacanthoma, amelanotic melanoma, pyogenic granuloma, lipoma, and adnexal tumours.

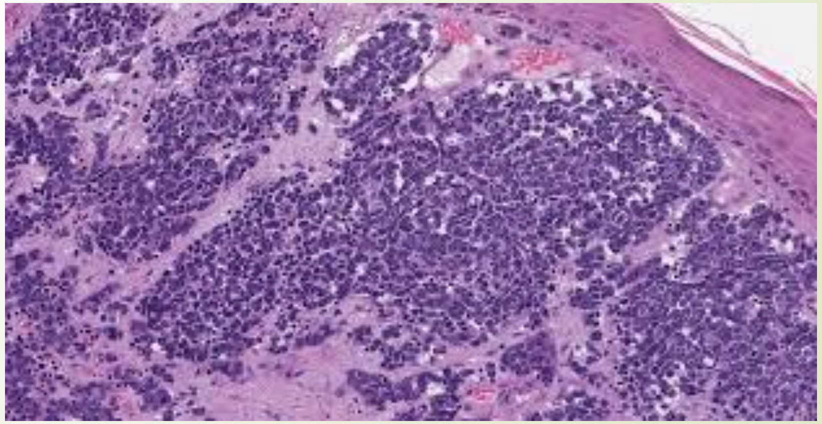
Histology can clarify the diagnosis in most cases. However, immunohistochemistry is required for a definitive diagnosis. MCC consistently stains positively for low-molecular-weight CK20, which is a fairly specific and sensitive marker for MCC, with a characteristic paranuclear dot-like positivity.

Reference:

- Tai, P. Nghiem, P, T. Park, S, Y. (2022). Pathogenesis, clinical features, and diagnosis of Merkel cell (neuroendocrine) carcinoma. In: UpToDate, Post TW (Ed), UpToDate, Waltham, MA
- Howell,R.S. Rice, J, A. Sticco, K. Donovan, V. Castellano, M. Gillette, B. Gorenstein,S. (2018). "An unusual presentation of Merkel cell carcinoma: a case report". J Surg Case Rep. vol(8), rjy185. PMID: 30093989

Merkel cell carcinoma

- Aggressive malignant neoplasm; most common in **elderly** on **head/neck**; also seen in setting of **immunosuppression**
- Erythematous to **violaceous papulonodule**
- Merkel cell polyomavirus (**MCV**) implicated
- Histology: infiltrative dermal/subcutaneous mass composed of sheets of uniform basaloid cells with high N:C ratio and finely **speckled "salt and pepper" chromatin** (usually without prominent nucleoli); **numerous mitoses** (often >30/mm^2) and apoptotic cells
- Positive stains: **CK20 (perinuclear dot pattern), neurofilaments** (perinuclear dot pattern), **chromogranin/ synaptophysin, neuron-specific enolase,** EMA, and CD56
- Negative stains: **TTF-1**, CK7, S-100, and CEA
- Main DDx is small cell lung carcinoma → immunostains very helpful!
  - <u>Small cell lung carcinoma</u>: **TTF-1+** and CK7+; negative for CK20, neurofilaments (lacks dot pattern)
- Diameter >2 cm, **p63 expression → worse prognosis**




THANK
YOU